

John H Clippinger "Workshop part 1: Introduction to Applying Active Infer

1:30:20

okay welcome back everyone we are here
in part one of the workshop that John
clippinger and our partners at first
principal first are coordinating so with
that for
the next session please John to you and
take it
away well thank you very much um I would
like to just to share a a deck with the
the group here if I could um and uh and
provide a sort of an overview of of um
what we'll be talking about in this
particular session can you see that now
is this is this good for
you okay great um and we'll be
anticipating with Andrew is uh sure will
be coming on and providing the tutorial
but what I would like to do is in in the
first inst of the is is being able to
provide an overview of why active
inference uh is important why active
inference agents this whole perspective
that we're taking here is worth people's
attention and pay uh and being concerned
about um and what it has to offer to
address some of the the the the
limitations of the current uh AI systems
for example large language models and
some of the uh agenda architectures have
come out of them um and and in that in
that context what I looking at is said
what is the current state of large
language models and and agent uh an
agents and I think you there's a lot of
activity that's been talked about but

there there's still no Mission critical applications yet there's a lot of an exploration you have people like um Salesforce developing Little Einstein that can do uh various kinds of uh workflow automation you have co-pilot and Microsoft but it's not it's not it's not really core critical not to say that it won't be but it is still uh is still on the experimental phase it's still a black box we do not know real lot how it works hence you have issues of data Pro Providence you have a question of how to govern it um and there's more recently there's a concern about you've invested a trillion dollar but there's an unknown business model coming out so there's a lot of money that's going into it but it's not clear that the current framing of uh of uh AI um as it's now known is really is going to be viable and part of the issue here is is that we and I love this example from The New Yorker is like who's modeling who we're actually putting data into the models we're trying to write representations and we have that interaction but there's not necessarily an abstraction that's coming out so it's this is kind of this uh relationship of of seeing ourselves mirroring ourselves and going and this sort of cycle um and what we have now is a certain hype cycle so it's around the peak of the this is 2023 from the uh Gartner group and it's the generative AI is at the top of the hype hype cycle and then if it's hard to read this particular uh graph but you'll see that

principle AI is is U basically um on the early cycle it's on the early phase of it but it's not seen as taking very long where some like uh you're going to have AGI is seen as being like 20 years out it's the top of the hyp cycle I think if we update this there's now I think it's starting to be appreciated that the generative models and the expectations and the kind of problems and the cost associated with them and the lack of improvement the the the lack of degree of improvement through successive models is starting to show through so the the Next Generation open AI shift strategy is a rate of GPI improveing slow in other words the new Orion model is not did not have the level of improvements that they were expecting I think that is something to take an account um and hence the cost of of training and inferencing is extremely high and and with that the the M the fact that you have this enormous consumption of energy uh that Microsoft is going to have to recommission Three Mile Island and drive one of their own data centers this shows that the Imp practicality of having to um have so much computation um in order to have work and make these models work and I I think that this is starting to be recognized within the the the limitations of large language models and sort of the the the parametric approach to to uh modeling uh AI modeling and I think just to start to step into what what we have now what

we're looking at I I love this is
another a cartoon from the the New
Yorker and since says to think it all
began with letting autocomplete
finishing our sentences and well
autocomplete is a sort of for Markoff
chain so we are we are having the sort
of chains on top of chains that we've
been able to develop some pretty rich
representations of of of human uh
activity text and actually forms of
cognition but there's not an underlying
model there and so what we're looking at
there may be some foundational issues
that are coming up in in very large
language models and that the fact that
it is a black box is not explicable uh
it locks internal causal models um
there's not a way of reducing the
complexity it's sort of a batch
processing there's no real-time updating
and that batch processing we seem is
extremely expensive and and so we we
we're caught on on on that that issue um
it's not atic it has no no no means of a
spontaneous adaptation and correction on
its own um and suddenly doesn't know
what it it doesn't know that has no way
of representing its completeness of its
knowledge and and adapting coherently to
that and and and also I would say it's
not it's not derived from sort of
fundamental scientific principles it's
really sort of an engineering phenomena
on a case-by casee base uh basis now
this is changing um and I think the same
issue if you start to see foundational
issues for multi-agent uh llms in other

words the transition to making
agentic architectures where you can um
create a particular agent that's
optimized to do a particular task or
role a lot of the success of that agent
is depending on the quality of the
prompts of that and sometimes those
prompts are good sometimes the prompts
are bad but there's not a really
principled way of doing that currently
and so they have a very fixed way of
of and specialized kind of goals it's
sort of a a a mechanical it's it's so
the instantiation of an industrial model
um that we see um now what we we rather
than having a a a sort of biological
self-correcting model it is s of these
fixed static models now what I would say
the active inference difference here um
is is that it's really developed from
first principles and I think and and I
think you're going to see the Andrew is
going to go into this much more detail
but it really C it cuts across different
disciplines that themselves have been
uring fairly quickly in the last two or
three decades and so it is going in
basic basic physics Quantum information
Theory the a whole idea of computational
Science Biology and cognitive science
all these are intersecting into creating
a new way of looking at what
intelligence and life are and how
they're
interconnected um and it also builds in
wellestablished statistical methods
including Markoff blankets blazi and
belief mod the stff the work has been

done with JD a pearl um and basically
what you at active infine does and and
just to highlight is it codifies a
predictive brain thesis that's been
nurturing in in neuroscience and so the
pr and then that in turn sort of mimics
the processes of the scientific method
and when I get so we think about what
the method is that you're you're trying
to codify and and I turn to Richard
Fan's comment well quoted comment that I
can't what I can't create I can't
understand so there's a new kind of
criteria for what it means to have a um
to understand something is be able to
model to create a digital twin if I can
recreate something then I understand
what it is and if I can replicate that
behavior and that that's sort of the the
the principles that you find in active
inference from the very beginning is ah
how do I grant a generative model how
does a living agent have a generative
model of it s in its world and this this
really owes it work his Origins back to
to uh the work that's done with J Pearl
on his understanding in causal reasoning
and also markov blankets you how does
some what is an autonomous thing and
this is where Carl fritzen anal work how
how does something maintain its
separation from the world in which it is
in and so it is not governed by that
things but maintains its autonomy and
there's a certain kind of topology there
that and something has certain kinds of
inter internal States it has external
States Observer States and action States

and there so there's a lot of deep Concepts that's in it I'm sure Andrew will go into that much more detail um and then what we have is is the active invers agents themselves as sentient live things so you're creating something and there very different than say a large language model you're taking something it has agency and it has a representation of itself in the context in which is trying to stay alive um and so it's self-organizing it's self-healing is self-replicating and symbiotic in the sense that it evolves complexity Mutual synchrony with other living things in its World in its Niche it's a joint Niche construction uh hence sentience is is sort of a is a kind of situational intelligence an awareness that's granted in the survival of that particular agent and the person that's sort of the the context of the person who's really behind this is Carl priston and and um my own experience of being someone who is very interested in sort of and artificial intelligence from from many many years ago and self-organizing systems as cybernetics it really hasn't been until the last five or six 10 years that this has come together in a coherent uh not only coherent Theory but a whole coherent formalism um and through what what car would call physics of living things and he's one of the most SED sists in the world it's a neurosci is has done um basically developed a reputation neuroimaging and infer the structure

from the brain from Neuro Imaging uh and he applied the free energy principle which we'll go into more discussion to later um and so one of the I love this idea of a tree foil knot it's sort of like okay is it is sort of this autocatalytic biotic system it's really important it's a living system and it has certain kinds of characteristics that are all intertwined with itself one cycle synchronizes is exception or it's a representation external world to minimize uncertainty the other does it internally it has to align that with the external world and then you have another cycle that has expected free energy minimization it makes predictions about how to change itself in its World um and so what we have here is basically a a a what I think active iners offers is a really a soundness of grounding in bi in biology and physics and so you have a sort of a ground truth of evidence space that do I make predictions and do those predictions allow me to act and survive in the world um so it can be explicit it can be explainable and and we'll see we'll talk about this later but you can have since it has cause of models and language and knowledge graphs it can explain why it doing what is doing and also it can talk about explain why how much it doesn't there degrees of uncertainty associated with it and there's also the potential of self-policing self cor acting mechanism and I think this this is really important in other words you can design a active

inference agent to to live with in
certain constraints and Bounds and you
can build in certain me me mechanisms
that have for self-governance that
cannot be captured uh so it's actions
are within the limits of its designed
Mission and not to have generate
negative externalities so this I will
sort of leave with this one and then
just to quote William Blake I mean who
talks about this

a poetic form is a fractal nature
reality see the world in a grain of sand
and heaven and a wildflower and hold
Infinity in the palm of your hand and
turny an hour this just sort of pulls it
together in a more poetic sense so on
this I would hope to uh hand off to uh
to uh Andrew if

uh thank

you I

great thanks John

um and so hi everyone I'm me Andrew um
Andrew pase I'm uh currently involved
with the active inference Institute it's
been great partnering with with uh Dr
clippinger with the director of The
Institute you know we've we've been
doing some really exciting work lately
um and I want to address a lot of the
points that that John has made although
most of that will come tomorrow when we
give some major updates on our biofirm
so today in the spirit of like an
initial workshop and kind of
introduction to active inference I'm
actually going to refer back to this
tutorial that I gave a couple months ago

uh at a International Conference for computational social science meaning folks who were involved in sociology economics uh you know a broad range of fields and uh the context was applying active inference a lot of these principles that John is referring to to social science research

uh and and I I basically make the the claim and the argument this is already being done so I'm not necessarily providing a bunch of new information but it's just to kind of extend the dialogue to computational social scientists like here's how we can apply active inference to you know your particular setting in different ways so uh we'll just kind of cover some of of what I went over and you know I apologize if there's any Whiplash from how quickly I go through these slides um but we're just going to move a little quickly um so the context again of this this uh this tutorial that I gave was it was split into three parts it was a 3.5 hour Workshop where we actually we started with talking about agent based modeling you know how it's traditionally been done and then followed by that something that I would call a cognitive turn in agent based modeling that's happened over the past couple decades as you know there are these kind of calls for you know we have the computational resources now we have an immense amount more of you know Empirical research and validation of different kinds of principles related to human cognition

coming from Neuroscience
neurobiology uh you know cognitive
modeling uh studies of neuronal Dynamics
uh organisms in their environment all
those kind of things so so there's a
shift to that and then uh the most
recent you know means of trying to
establish that the broader
state-of-the-art
area is uh using reinforcement learning
agents so we'll touch on what those are
uh and kind of how that work works and
you know anyone who's kind of been
looking at the the agentic uh framework
for for what's being done today they'll
probably already be familiar with
reinforcement learning then after that
we'll go into active inference and how
that kind of kind of bridges together a
lot of these different things um and
apologies I want to just mention real
quick uh you can actually go to this
this link uh the quickest way of
following that is just by going to
Google type you know my name Andrew pase
and then you could type GitHub or ic2 S2
in any case you'll come up to uh the
landing page for this you'll see links
to all previous slides I've made uh
slides that I'm making for tomorrow uh
I'm also making a lot of his Open Access
uh so so there's code that people can
you know play around with and things
like that um so agent-based modeling uh
this has been done you know this goes
back to roughly 70s 80s with a lot of
you know uh initial developments of
computer technology and being able to

realize that oh we can we can simulate
uh people right in some loose sense in
in some kind of simplistic sense uh how
can we simulate people and why would we
want to do that we would want to do that
so that we could maybe imagine uh
hypothetical scenarios imagine you're
designing some kind of policy or want to
test a policy uh and see you know oh if
I recreate that in a simulation and have
people in the environment where I'm
implementing that policy uh what happens
what's the collective outcome you know
what occurs so all kinds of fascinating
work from uh shelling and the the
traditional way of looking at uh housing
segregation for example th handling
those kinds of problems where do people
land where how do they settle you know
in in in uh like a residential
environment uh even more interesting
work later after that Santa Fe Institute
creating an artificial stock market uh
where you where you have agents who are
basically interacting with this kind of
financial environment where they're
looking at stock prices changing they
make choices to you know buy or sell Etc
and see how that plays out Recreation of
housing booms and busts excuse me not
housing Financial so this can be applied
in many different ways point and so what
what is agent-based modeling you know in
its fundamental
principles and how does that relate back
to active inference which we'll get to
um key components of agent based
modeling is that we have agents kind of

the catch word of the day um
traditionally agents are you know that
they just represent some entity a living
organism you know this this can be used
for for biological and Zoological
studies trying to understand you know
how ants or or insects you know
interacted environment or at a broader
scale how humans and entire you know
societies and cultures might interact
with their
environment and and so what's really
important about this with agents
environments simulations is that know we
just kind of run a simulation over a
series of time steps what and so we let
Dynamics play out and then what is it
that that composes those Dynamics it's
the assumptions we make about agents
what is their behavior what do we think
that they do you know how do we program
them in this way or in that way uh and
then also what are the aspects or
characteristics of the environment you
know and so again given that a lot of
this work was done starting in the 80s
going forward with with with very you
know we nearly archaic at this point uh
forms of like computational resources
it's like things were kept as simple as
as possible and there are a variety of
guidelines that need to be used to
develop agent-based models for running
these kinds of simulations uh you know
and so so some of them even running up
to to 2007 here this is the work of of
doctors laser and fredman who are
respectively uh have been involved with

Northeastern University in Stanford um
you know that they give some guidelines
for how to do this um guideline one and
two uh in some is you need realistic uh
assumptions that you're making about
your agents but not too realistic in the
sense that they're too complex one
because we don't have computers that are
Advanced enough to run highly
sophisticated complex agents and then
also you know if we make if we make too
many assumptions then the situation
won't generalize well you might be
making an agent based model for one very
very particular scenario where it it's
not adaptable across different real
world circumstances right so we need
some kind of balance in terms of the
complexity of the assumptions we're
making the Simplicity of the assumptions
we're
making the rest are a bit more plain
which are one uh reproducibility you
should publish your code you should
document it very well other researchers
should be able to kind of take up the
torch and be able to experiment with
your code um you know we at the Active
inference Institute we're constantly
putting out new get Hub repositories and
documenting or processes so that other
people can come get involved so I just
you know want to mention that there's a
big problem in reproducibility in
science today and it's very well known
you know so uh just wanted to touch on
that the nont reality it's like why do
agent-based modeling uh if it doesn't

help us uncover anything you know why
why not just sit down and have a
conversation about you know what what do
we expect to happen what are our
thoughts about what's going to happen
why you know do all of this um with with
setting up these programs and and agent
based models and designing them if we
can just come up to the same conclusions
through speaking so you you want your
results to be interesting you want to
make realistic assumptions that you know
we can kind of have all these things
synthesize um together so I made some
points about balancing Simplicity and
complexity this is just a representation
of uh you know this these These are
common problems in machine learning and
deep learning you know you you go and
get uh you know University education
these are the core components of all of
your courses you know this isn't
specific to active inference like wide
ranging you know concern how do you uh
create a model that is realistic enough
capture uh and and explain the en the
phenomena you're looking at um but uh
simple enough to where it's adaptable to
different circumstances you know there
you know these are kind of in part uh
you know just as uh Dr clippinger
referred to Richard feeman you know I
just some reminders from you know famous
statistician George GP box you know all
mod all models are wrong but some models
are useful right we're always trying to
approximate a reality and so the
question is how do we come CL closer to

approximating uh our reality and
so how do we actually you know using
modern computational resources using all
these new advances in all these fields
how do we actually make you know new
forms of what we'll call cognitive
agents who exhibit the kind of mental
Dynamics and social dynamics between
them that actually better approximate
real human
behavior so I give some definitions the
differences between traditional agent
based models and what agent assumptions
are are being made and uh more often
than not for the sake of Simplicity an
agent is pretty one-dimensional if they
observe X then they do y otherwise they
do see these very simple pre-program
roles they're not autonomous they're
they're kind of like you can think of
them as like thermostats or something
like that right the temperature is too
high you you turn it down if it's too
low you turn it up um that kind of
reactive framework uh there's no beliefs
about what's going on there's no
internal goals per se aside from a
single homeostat that is you know
maintain the temperature at what you
said um cognitive agents however the
expectation for those this throughout
the the recent uh agent-based modeling
literature is like how do we achieve
cognitive agents we exhibit internal
cognitive mechanisms they exhibit
perception action they actually learn
and update their models I'll make a
point that you know as John referenced

with in the case of llms which I'll cover much more deeply tomorrow llms do not I mean it's so expensive to train them and you typically just train them once right and so um they don't learn in real time they're kind of stuck with the knowledge that they have they have a lot of it to be fair they've been trained on you know thousands thousands thousands of documents uh and so on um but but that's it like that's you have and then and then we have to figure out how to work with LMS from there with cognitive active inference agents as well as other kinds of cognitive agents the ideas that they can learn in real time meaning in the moment and adjust their expectations their beliefs to what's actually going on now um and then also planning so the capacity for being able to infer or guess using evidence that they take in like what's going to happen in the future right and that's immensely important in today's climate right where we're that I mean that's in part the reason we're doing in face model is to attempt to plan to test things to to come up with our own inference about what's going to happen in the future and then act in actionable ways to to realize our own goals in the future right so that's that's you know ideally that's what an agent would do and be capable of um and then with you know this cognitive turn we we're also looking at a broad synthesis of all these fields you know as John mentioned we're looking at things like cybernetics

and machine learning we're looking at physics information Theory um people are starting to it's great it's it's fascinating that so many uh interdisciplinary conversations can come up and collaborations between researchers of various backgrounds because it's like oh you know once we once we realize what it is we're trying to do here we can start to pull from maybe principles or relate principles from other fields to this field what works what makes sense what's empirically you know what can be validated so on so it's it's um you know it's a very rich field right now and then furthermore a lot of traditional agent based modeling has been done using uh uh software called net logo which is very specific uh it's it's good if you want to do agent based modeling like you just learned this one programming language there's a GUI like an interface for using it highly simplistic it is not integratable with other tools not integratable with you know llms with natural language all those kinds of things um and it's so it's it's very narrow you know it's good at what it does uh with cognitive agents we would want to be able to pull from all kinds of other resources that are you know available including open source ones uh being able to use apis being able to pull in other kinds of real world data not just purely simulations you know so um there's just a lot of richness behind this idea of moving to cognitive agents

then there's some basic you know uh ideas of what makes a cognitive agent again perception they take in observations and they infer what's going on in the world and kind of like we what we do my stomach growls and I infer that I'm hungry for it's a very simplistic example um and then agents are are generative the sense that they actually come up with uh their own plans or a you know choices of what actions to take based upon what they believe such as I observe my stomach is growling I infer I'm hungry I will now choose to go eat food as like an affordance that's available to me in my environment so this is a new way in a certain sense of doing agent based modeling where before we were looking at modeling an environment with a bunch of one-dimensional Agents you know predictable agents on the inside if they see X and they do y otherwise they do Z but now we have models in a model in the sense that we're actually able to internally inspect our agents right by having this beliefs based frame framework where agents observe real data or information and then infer beliefs about it we're able to then retroactively look inside the agent and see what its beliefs are as opposed to these kind of you know very difficult to interpret uh blackbox models that we find very commonly in deep learning and and you know very very highly performative yet very uh uh difficult to interpret like neural networks and

architectures like that with with uh
agents we would want to be able to see
in a in almost like a human sense like
what are their beliefs that they're
holding and why are they doing what
they're doing so we need to be able to
accomplish that in order to actually
realize all the additional benefit of
doing this which is being able to look
at you know internal cognition what are
agents learning whenever I whenever I
implement a new policy or something
right I want to be able to know what's
going on with them internally I want to
know the full impact of of what it is
you know this plan I I'm considering
like
actualizing so let's bring it back um
you know how how do we address all these
traditional criteria how do we make
realistic assumptions how do we um you
know how do we make our code
reproducible that's an easy one because
like well we're still coding the whole
time uh and then you know how do we how
do we derive useful insights from this
um and what are other problems that are
going on like now that we're getting
into cognitive architectures
um and and we we are starting to brush
up against the potential of recreating
more blackbox models right we need to
keep things simple enough like how how
do we navigate all this a common problem
in in social science research and
general but also in data science machine
learning and those kinds of fields uh
reinforcement learning common issue is

what is h how do we handle What's called
the explore exploit tradeoff this is
this is famously an issue uh if you
have't agent who comes up with a
suboptimal uh solution to a problem you
know let's say an agent who's you know
they're trying to figure something out
they find something that they deem to be
good they get a positive you know reward
signal or positive feedback from that
but it's suboptimal there there's
actually a better solution exists they
just don't know about it yet how do you
get an agent to actually be willing to
explore same thing for humans right we
we don't always stick with the you know
the first decent looking you know option
available to us sometimes we we explore
in the sense that we seek better
solutions to problems and this is
especially important and something to be
addressed in agentic Frameworks where if
you're trying to use agents for say
software application or use case like
you want them to be able to find better
Solutions ideally you know at any given
moment especially in a changing context
that might call going forward for better
Solutions you know say agents who were
maintaining an environment that's
subject to uh natural disaster or or or
or climate change right like that that
might call for a whole new set of of of
policies or actions that need to be
taken uh you know that that couldn't be
relied upon in the past so we need
agents who seek new
information um unfortunately in

reinforcement learning you know that this is still an issue um people are trying to figure out how to kind of manually Define what what defines exploratory behavior and it tends to boil down to kind of simple mathematic of of defining like okay well we'll s we'll we'll modify these agents so that 90% of the time they go with the best solution they have they know about and then 10% of the time they'll just pick something random that way they're at least exploring 10% of the time you know you could put it that way active inference again I'm sorry I'm kind of zooming through a lot of this call this is over three hour tutorial um but uh rein with active inference ages this kind of gets naturally naturally resolved in terms of What's called the free energy principle which we'll be getting into um but basically the idea is if you're able to to have agents who naturally seek information perhaps in line with how we do right that when they come up against something they see oh there's value in seeking new information to find even better Solutions the ones I already have and being able to to do that in a way that isn't some kind of ad hoc rule of you know the 90% versus 10% as I mentioned that's that's um kind of a key component you know a key consideration designing agents um so this brings us to the active active inference which I I describe here as a framework for doing agent-based modeling it uh begins with the free energy

principle uh that I I'll make a lot of mentions to this nice textbook that Thomas par Carl friston and you know their other co-authors published a couple of years ago um but you know there perhaps even some more updates to be made because it's such a lively field but uh that textbook really lays out a lot of the core principles and and ways of getting started with active inference um the the free energy principle everything that changes in the brain must minimize free free energy now despite all of this energy disciplinary communication dialogue between you know again cybernetics with psychology with behavioral economics um that actually sounds quite simple right to be able to boil everything down to oh we have a brain it's trying to minimize free energy that that that almost sounds simple enough to where we might be able to employ that as a kind of you know simple assumption that we make an agent based modeling it's quantifiable uh it's it's a presume Pres assumed is the wrong word uh but but something like a principle that can be applied in any given circumstance whenever we're looking at you know the these organisms who are self-organizing such as humans such as you know living organisms such as you know plants in an environment that's being taken care of uh you know that they exhibit this this kind of uh characteristic of minimizing free energy as a as a form of quantifiable

uncertainty so we'll touch back on that again in a moment but the the the what active inference provides is a kind of normative holistic framework for what I argue should be applied to agent based modeling it it itself addresses agent based modeling that phrase is used in the textbook this is not a foreign you know out of left field kind of invasion of active inference and agent based model and proposing there's so much overlap and I think it's just an underutilized connection that's already been made right um active inference is based in you know it it provides neural process theories meaning that we're not just you know assuming in the same way with agent based modeling we might assume oh all people are utility maximizers um in this case we're not just assuming it we're we're looking at neuronal Dynamics we're looking at you know Decades of research into using neuroimaging studies and looking at how you know how does the brain really function how does it handle uncertainty different kinds of Behavioral Studies and tasks where you can you know record uh MRI or EEG along the way and look at the Dynamics of those um uh of that data that's being collected along the way and um and then from there we can start to kind of model how that works so um as I already mentioned uh we're still looking at a kind of belief-based framework which is to say uh you know we're looking at uh for for agents who have beliefs and they update

their beliefs and they do that by way of the free energy principle which is minimizing uncertainty uh we we start we start to ex we start to see these organisms that we're modeling humans or otherwise we start to have naturally emergent adaptive Behavior right by by trying to figure out what's going on in your environment and then adapting your actions in order to better understand it to better realize your own goals you're effectively minimizing your uncertainty and uh the free energy principle basically is a kind of free energy is like a proxy for that uncertainty that that renders a lot of these things computationally tractable meaning that we're able to quantify it we're able to report it whenever we run agent-based modeling we're able to like actually look on a chart and a graph how much free energy is being reduced uh over time you know and and look at the agent internal beliefs about you know what what is it that they're figuring out uh within their environment over of course is simulation um the view from from this perspective is that uh another aspect I I already addressed the expl exploration exploitation balance and tradeoff uh self-evidencing this means that agents in order to minimize their uncertainty um one way that they can do that is is um to try and act to realize their goals there's something they want something that they expect right and so it's as if you have this prior set of beliefs of what you want to see in reality this is the

you know it's this is the kind of framework that active inference deals with but also referring back to you know Jude Pearl as clippinger mentioned uh you know this is a kind of basian idea you know this is rooted in something like like basy and uh statistics or even mechanics where the idea is that you you have your prior knowledge you bring it to the situation and then that will impact how you update your beliefs based on new information right so we're we're not always starting from some blank State some some starting from scratch we're always coming to a situation with what we thought before and then that gets updated how does that get updated from active inference that standpoint it's through free energy principle um so you know it's I made some points in this tutorial because I realize we're we're we're really uh uh you know I want to make sure we have time to to kind of bring this back um but uh with active inference that we already have so many aspects of uh reinforcement learning and active inference which overlap we're trying to create cognitive models you know active inference I don't want to say it's necessarily competing I don't know if that's the right word but it's you know it's it's making advances that reinforcement learning is trying to make itself so um and uh and with active inference it's like well we with Act inference we're actually basing this on you know a variety of of principles

you know

that again sorry zooming through you
know we're able to like cognitively
model you know what's what's going on in
the mind in various ways that
reinforcement learning is really lacking
um today it's it's really lacking that
it's it's uh you know these models are
being built to be highly performative
but they're missing the mark whenever it
comes to something like realism and
basing

uh basing these reinforcement learnings
and something like a you know science
that extends Beyond just the mathematics
of machine learning right we we want to
be able to find biological physiological
substrates for how all these things
happen right um we want to be able from
there to be able to apply everyday words
and interpretability so what is it that
defines what a habit is what what
process is involved whenever a goal is
formed and made and then from there uh
you know attempted to be achieved
through action uh you know what defines
planning what is

prediction um so I just want to see if
there's anything else I might be able to
touch on this was a nice um you know for
for for anyone and maybe I should have
started with this but this is a little
bit more of an intuitive glance and this
is actually these are not words from uh
Carl friston surprisingly uh they're
from uh Sarah Feldman Barrett who's at
Northeastern he uh you know nearly as
rep reputable as Carl friston in the

Neuroscience field and I just personally
uh as someone who's also interested in
in in neuroscience and cognition you
know I find her work fascinating she's
done a lot of work on um emotions and uh
you know kind of affective Neuroscience
so it has to do a lot with looking at
what's going on inside the mind of you
know especially humans and so this is
kind of trying to provide an intuitive
glance that you know how can we think
about B in Frameworks about um um
inference about active inference as well
so it's like uh this is a long quote
from a podcast that she did your brain
receives sense data from the world
sights and sounds smells and so on
receives sense data from the body so
when there's a loud bang or a tug in
your chest your brain receives that
information as the outcomes of some set
of causes but what caused that loud B
what caused that tug in your chest your
brain actually doesn't know it only
receives the outcomes it has to guess at
what the causes are this is called an
inverse problem you know what the
outcome is you don't know what the cause
is so your brain has one uh you know
another source of information available
to it that is past experience memory um
apologies for just like doing a
slideshow karaoke here but you can read
it for yourself I suppose but the point
is and with some of this highlighting
it's like we're actually taking account
of a variety of things you know we're
taking account of interoception meaning

observations that arise from within the body or for an active inference agent uh you know it it it it has to kind of model itself you know I mentioned earlier you know if I if I'm hungry this goes beyond the five senses in that kind of perspective right touch taste feel all of that um with this it's like well I need to understand what's going on within my body I need to understand what's going on outside of my body as clippinger you know referred to earlier the idea of a Markov blanket it's kind of like I need to figure out what defines the boundaries between myself and what's around me you know on the one hand I need to have a model of myself in order to maintain it uh otherwise I succumb to the environment around me to use some of the language that's used in uh you know the textbook it sounds a little Bak but you know we're referring to real world circumstances if I you know if I don't take care of myself I will naturally Decay like that is kind of the sort of the principle the world is entropy so to speak right it's like if I don't eat food you know I I fall apart so I I need to have a model of myself what's going on within I need to have a sense of what's going on without uh and then and then I need to be able to kind of synchronize those things in such a way that I can continue to exist as an entity within this environment around me that's a that's a that's a very intuitive you know perhap perhaps not very well technically described but

for now an intuitive sense of of kind of
what defines a markup blanket so you
know for this is a sort of foreshadowing
for tomorrow we'll talk about uh this
this really interesting project we're
working on for the biofirm uh with with
with uh you know Daniel and and and
under guidance of of John um you know in
that in that sense we're we're trying to
develop homeostatic agents uh
homeostatic meaning agents who you know
their own sense of their
survival is predicated on uh the the the
homeostasis that their environment
around them so we're trying to create
agents who
take care of and attend to the uh the
environment around them as if their own
Survival depended upon it you know this
is a very interesting kind of uh kind of
flip you know where where where you
actually want to match and model the
environment around you but also take
care of it in a way that you can
maintain it within those bounds just as
you know I have to eat in order to
maintain my energy levels you know that
the agents need to take care of their
environment you know within some kind of
range to where it has you know enough
caloric energy so to speak um so um I
think for now John is there anything
that you think I have not addressed that
might be useful to to our audience or if
anyone else would like me to touch on
anything maybe something I skipped or or
otherwise i'd be happy to to speak
more no I I think you you you've uh

covered some of the really key points that um that are that what I I think is you've done is what differentiates active inference from the more traditional agent-based modeling I think is important because when people think about agent-based modeling they they sort of default to what the traditional models are and I think that you've done an excellent job of sort of uh highlighting what the difference is and what the the the having having that broader perspective that more grounded interdisciplinary perspective you you're able to more not only represent the complexity of it but also provide a way of modeling agents in within that complexity um and I think that the the the the notion of a the the homeostatic nature of of of an agent trying to live in in conjunction or really harmony with the world around itself is depending upon its survival and its ability to stay alive and differentiate and have complexity but the synergy of the environment is really another key point because I uh I think a lot of people they think oh well you may say like navigating a fitness landscape you're just trying to optimize different levels but in fact the landscape itself is shaped you're doing joint Niche construction so it's a much more complex notion than traditionally understood General modeling and I think in in consideration we looking at tomorrow this whole idea of the basic character of model having it

based upon principles and biology and nature and having being able to create an economic agent that acts that way I think is is is really quite important um and so maybe some people have some questions about that that that we want to to focus on or not but that I I that's something that that I think it's important to highlight

yeah awesome presentation Andrew Ernesto if you have a question also anyone in the live chat is welcome to ask a question no questions thank you very much this is so inspiring well and also just to to put it in context I know this is something that you're very interested in in this whole idea of a btic and economic system and been looking at this yourself so this is really trying to lay down the principles really the the scientific principles and computational principles of doing this and and and uh and we'll get in more to that tomorrow but that that's the the intent

here yeah a lot more to say again in our remaining 40 minutes I'm happy to read questions from the live chat I'm also happy to screen share and look at some of the registered participants backgrounds and interests sure okay okay you can unshare Andrew and I'll pull up some word clouds all right okay okay so let it

load okay these are word clouds that are generated from the participants registration information and we'll we'll zoom in on them and and uh I'll I'll just kind of mention them and we'll we'll zoom in and uh

if anyone has a remark or something that they want to uh like highlight or call out about it just go for it okay so this is a word cloud of the uh 150 or 200 registered presenters overall asking about what is

your personal background so the biggest and and and the this is scaled just by number of uses of each term so one person mentioning a term a ton of times might make it large so just to be taken as sort of a representative we see those tiny two letters AI which could mean almost anything at this point and a lot of other terms like computational research systems

Neuroscience

cognitive philosophy social

academic robotics technology

practice this kind of speaks to the broad range of topics that that that leads people

into joining and at least entering that information on the uh form any thoughts on this or we'll keep on exploring these word

clouds okay okay all right again just just raise your hand or or or go for it interest

okay well there's I'm interested the notion of community and systems I mean

applications the community seems very strong yeah yeah and that that's the the sensor of a of a later one so here we asked what what would be useful and by asking what would be useful and what would be interesting we alluded a little bit to the Dual composition of the expected energy what would be useful what would be utility what would be pragmatic and then what would be interesting what would be epistemic so hearing what would be useful you know here we are in in applied active inference Symposium wondering about how does active inference go from being a unifying framework for ecosystems of shared intelligences and all these different kinds of things that we've heard already and we're going to hear in the Symposium to come so interesting that systems a way to operationalize and really move forward from framings of complex adaptive systems to having the implementations and also the community and the people understanding research models collaborators and then a lot of different topics that are being explored in more mainstream computational science Fields like multi- agent algorithms Andrew alluded to that and kind of brought that up in the context of the the Goldilocks parameterization question the agent based model can be so simple it gives you a great intuition it shows at a a demonstration layer how

simple Behavior can lead to complex
outputs on the other hand the agent base
modeling can go the other extreme where
it's so sophisticated then it loses its
transferability so that's what people
would find useful and then
here's what they put for what would they
would find
interesting
applications make sense as that's what
this Symposium is all about and again a
lot of other ideas thinking about
different specific systems research
ideas architectures synergies toolkits
insights
implementations
support Collective
directions here's
asking what would help you in your
learning and applying active inference
and community
looms
large other people as well as some of
the more tool resource textbook
education examples Frameworks
interactive
tutorials different languages
python what does anyone think about
this I'm uh a little surprised that we
not seen the phrase llm anywhere
um I just uh kind of noticed that so one
of the one of the points that that I you
know have been trying to work out and I
hope to speak more on tomorrow is um you
know integrating active inference with
llms and uh you know there have been
attempts to to do this and even even
with priston and Par and folks you know

that there there's been attempts at trying to pull out an agent's beliefs and then use an llm to process those beliefs like kind of decode them and re-encode them in natural language so to speak it's as if the agent is speaking and saying what it thinks and believes and why it did what it did is you know another layer of interpretability and uh I don't know I'm slightly surprised to not see llms anymore um you know it maybe it's a little too maybe that's a little too on the frontier right now um I know it is in many other areas you know and it's um sorry in the couple minutes we have it just uh you know something that I've been working on personally and this is feeding into the the bio firm is um how how do we how do we use llms in a way that we can actually integrate them with you know all these major priorities that we have uh where where we want to have agents who actually have defined goals right and and and ways of trying to stay alive you know in a way that they interact with their environment in a way that it's as if their own Survival depends upon it we don't want llms which you know again I'm just saying this quickly you know I don't me to sound disparaging but llms in one sense can act as like functionally as Talking Heads where you just send in some text and they try to predict what would be the best text in response they still subject to hallucination to confabulate making things up so to speak even though

the answer looks structurally sound you know and and and so the my point is that rather than viewing llms as uh as agents I don't I don't I don't think they are I think that uh you could have an active inference agent who is augmented with the capacity for natural language by having it interact with an LM right so yeah sorry to so deep my own yeah scan scanning across all the registration fields for llm mentions so how does active inference apply to the design of llm systems AI outside of llms better ways of Designing internet skill protocols and governance systems and should converge with tiny ml to challenge llms deep learning so it's it is interesting that that it's uh referenced with only a glancing blow and and uh We've even seen earlier like in the responses to similarities and differences with reinforcement learning with friston earlier about the increased amount of attention and Tool development and the later and more professional maturity of training environments and this is like one of the key fosi for the open source active inference ecos system and for the Institute is to develop those kinds of methods and trainings and all on boardings all these different aspects of of increasing the professionalization so let's look at the challenges so this was what what what are the challenges facing your learning and applying active

Influence People interesting the problem
problem and the solution models the m
math
software maths again mathematical
understanding difficult physics
scaling simple
examples you know it's it's it's
interesting um because in the larger
conversation about what AI is I mean
it's it's that that conversation is so
dominated by uh large language models
and quote generative models within large
language
models um and and what least I've been
observing I tried to refer to my my
initial slides is that they're starting
to appreciate the
limitations of of the large language
models moved towards more agentic models
and more towards causal models and hence
creating aent architectures that do work
with large language models but they're
still there's not they're still not
based upon biological principles that
we're discussing active inference there
more of the mechanical models being
applied over and I I think to me from
from a point of view of sort of the
um the sociology of science soci it
there there seems to be these different
worlds that that I mean the number of
people that i' I've known sort of in the
open AI well they're they're very
unaware of active Ence and some of the
critiques that are if you if you listen
to the Gary Marcus he's got a very
pointed critique of of um of of large
language models and not being able to

achieve abstractions but that doesn't extend to appreciation appreciating what the the opportunities are an active inference and so there are there s of these islands of understanding that they still exists

um and I think that's one of the challenges and when you talk about developing an infrastructure I me you need there's huge financial resources are going into the the the large language models

infrastructure and where I think that the totally bias on this I think all the the opportunity is totally within the active INF of free energy principle and I think that's where the science is going and yet it's it's still it's sort of in in this isolate World um and I think the only way to get out of that is start to to really start to showing the applications and showing the with results and and uh because uh but it it's it's it's interesting I me to see they're they're different they're different worlds um um but I I think that what's the there's a peing and so the hype curve around uh uh the large language models and and it's going to they're going to descend my bet is they're going to descend into a valley of doubt and you suddenly you're going to see that showing up and the in the Nvidia stock going down and then people are saying you can't you know destroy all the natural resources on run these models you don't have to you know the other and Carl alluded to this in his

earlier comments today you know I mean he was talking about climate change and models compression I and I think that's going to be a driver of this um so but that that's some my my observations on on these yeah these clouds is yeah I'll bring in a question from the live chat also welcome Ma just thought I'd bring you on if you wanted to participate in this panel or just chill so Stephen selette wrote is the lack of llms mentions because large language models are different to Dynamic agent belief modeling and you want to take that you could you could you repeat that uh yeah question is the lack of llms in the word clouds here but just more more generally thinking about similarities and differences between active inference generative models and llms is is that because large language models are different than Dynamic agent or belief modeling I I don't want to approach this this question in the in the wrong way but I I will say that that that as far as you know whenever it comes to llms and and dealing with you know they internal parameters are obviously highly quantitative and then they're dealing with with natural language as to and then switch back to looking at active inference agents which you know are highly quantitative in nature and uh typically you know if you're making continuous time models uh you're looking

at again quantitative information and then even with discrete models you're looking at you know these kind of discrete values that you could potentially move in the direction of trying to you know make it I think this has already been done uh but uh you know there there are agents who um you know have been used to try and understand like oh we can read individual tokens then move up to another layer and then we have entire sentences and then move up to another layer we have entire you know paragraphs or ideas that kind of thing is done but I think this kind of like llm with quantitative uh grounded agents like that connection uh is still that in my view that's kind of the frontier of where things are at and so it's difficult for people to conceive like how can we how can we Translate encode decode re-encode natural language in a way that's readable by these quantitatively oriented agents and then vice versa how do we transition quantitative beliefs into natural language that's the very thing that I've been uh you know looking at thinking about uh recently I made like a active inference like PDP agent who uh you know they what they do is that they use their beliefs about which action to take and then that action that they carry out is actually corresponding to a prompt to an llm and you can actually take those beliefs the quantitative beliefs find some way of meaningfully interpreting

them and then putting that into these blank spaces in the prompt is as if you can have the agent say in real time I currently believe this with probability this give me you know a solution to this problem you know based on so so being able to find this kind of newer versus the quantitative and the qualitative you know it's it's historical problem when working with with data that go back centuries at this point so um it's it's really exciting though it's really exciting to think of how to bridge that kind of Gap and I I still haven't seen anyone you know nail that down this this is the frontier this is where we're at and I think that's kind of the next thing for everyone to really be considering thinking about working on um and I'll give my view on how that could be done

tomorrow

well um yeah I mean effectively llms don't really encode explicit beliefs or really maintain a persistent model of uh the world um even when we sort of give them a sort of loop by which we um communicate with them that Loop in itself doesn't actually factor into the predictions that is going to do um so they they just lack a framework for recursively updating beliefs which is probably why it's not being thought of here um but also mainly that you know llm agents are only agents by virtue of sometimes being connected to stuff and having um explicitly stated goal but I don't think that actually qualifies them

to the
agents you know yeah I I totally concur
with that I from my perspective I I
think that the reason
that LMS are are are not looking at that
are are not looking at or that whole
mindset is not looking at at belief
structures in in subjective experience
is I think it's a product of a whole
sort of objectivist reductionist
methodology that um is has been tuned
not to look at agentic systems I mean in
my generation if if you if you were in
biology you and and you talked about
agentic systems you you were you were
outed I mean that was that that that was
verboten and I think the the there's a
whole school of thought there's a whole
set of of in aonian sense a normal
science built around uh a sort of
objectivist reductionist non- agentic
notion and and and uh and so there it's
a new generation that's coming in in
both in biology and in and modeling and
cognition there's recognizing this
notion of an agentic perspective of
beliefs and updates and and that's a
huge shift I think in in sort of the
epistemological perspective of of a lot
of what has been traditional
science that's really interesting
there's also just
operationally the difference between
taking a very large fixed number of
parameters in a given model 70 or 7 or
405 training on existing data and then
finding patterns or igen modes or
activation um summaries that have

secondary interpretability but there's not really an effort of trying to find interpretability at the level of activation of a given neuron it's about looking at patterns and trying to find Collective level interpret abilities like in in the AI interpretability field contrast that with designing from the ground up an example agents where the beliefs are explicitly encoded numerically and so that's a different mode of building and also one that's hinted at in in this 2023 paper with Ma and others that highlights element of explicit semantic interpretability and also explicit semantic interpretability of uncertainty estimates whereas a string of tokens emitted by an Llm that says I'm not sure it's sure about saying that but that doesn't mean that it has a given certainty about the reference so understanding what the communication semantics of Llms are is incredible and what they do and I think when we look at this Cloud for people's responses how can active inference make a difference in 2025 where there there is Llms here many many kinds of systems and that's sort of like a microcosm of what the Symposium is different people with different systems of Interest asking H how do these different attributes of the model that we're talking about the first principles grounding semantic

interpretability surprise and
uncertainty and its bounding as a
central component as opposed to reward
how do all those things really play
out what does that demonstration look
like in in different
systems h

here's go for

Andrew no I was just going to say on
that that that previous one you were
just looking at I I I really appreciate
that kind of a broad-mindedness that
that people seem to be exercising I like
seeing terms like uh symbology scientist
governance um

industry I mean those are some very
very excited to see people who who want
to you know so much work has been put
into kind of the theoretical and then
sort of like in in in vitro and and in
lab and and in simulation kind of work
that's been put into active inference so
I'm just really excited to see you know
with the the um

International the the conference the the
back of inference conference that was
given recently and being able to see
like a drone that's flying based upon
the free energy principle and things
like that I'm just really excited to see
more

kind of real world applications right
and and kind of uh creating uh you know
systems and and applications for like
real world use cases it's it's really
cool to see some of those terms
there yeah I think that's going to be
really critical so see

yeah here's the cloud for how are you applying active inference again systems I I think just reflecting people talking about variety of different systems types but and also reminding me of my own journey to active inference being excited about complex adaptive systems however not really having a software toolkit or a formal way to proceed past okay we think about nested multi-agent systems and then where's the software toolkit and then we're right back to where Andrew showed us on the slide it's like well we can do a really simple one in net logo or we can engage on this Enterprise scale engineering research program and yeah here's a more incremental interoperable way to have portfolios of motifs and generative model components and be assembling and scanning across these different kinds of compositions like three ant nestmates in the desert three ANTM nestmates in the forest and these different kinds of environments being able to compose them and over the last several years seeing a lot of the advances in the category theory in the first principles physics surrounding all these sort of research things and it's like that's kind of the water drawing out and then that's the question about how it looks when it is applied and that's what I guess we convene around in this Symposium and and are seeing differently in 2024 than in previous years that's curious so what what's the change that you've seen in pre previous

years versus this year and then so
looking forward more of an
application Andrew or ma what what what
do
you think I was yeah
but absolutely that that that is I oh
yeah please yeah I mean I feel like um
the work of Maxwell ramstead C Casper HP
I mean obviously Carl but um a lot of
people who perhaps came from more
Dynamics uh decided to take a real
interest in what we could do with
something that was so scale-free and I
find that it's been really really rich
in terms of the different kind of
theories we've tried to sort of reforme
I know that's what most of my work is is
a Reformation of existing social
sciences um and to your point Daniel the
the problem we still have is that we're
missing like these key little plugs um
that yes you could couch as you know
just different schemes of message
passing but you still need some
interesting tools to to have a a very
good description of the vast vast
variety of types of structures that we
have Which social sciences have done a
really great job at if not intuiting at
least describing properly but we still
don't have the best tools to make
accurate predictions and that's part of
the reason why most of our polls when we
make them are are n they're useless and
that should be the main thing we apply
them to is just literally just
understanding where we're at not even
making predictions just understanding

where we're at and we still fail at that
so I'm very
I'm very encouraged by by seeing this
and I hope that we will start valuing
more and more interdisciplinary
perspectives that are sometimes more
technical but also sometimes uh you know
more philosophical or um or
social awesome also while we're in this
in in the coming workshops tomorrow will
be more of a Biore Regional biofirm
focus and then day three will be
entirely with
at all on this topic and and I want to
show one really operational open science
Crossover with llms and active inference
which is and I'll put it in the live
chat as well here I used
perplexity which does an internet search
summarizes and translates information
using llms so here we'll pick on Alexi
there's a a profile a learning plan and
a set of small medium and large project
proposals about active inference so this
is just summarizing expertise and
experience here we have a learning plan
it's just a starting position but this
brings a new low bar to the kinds of
resources that everyone can take on in
their learning and use as like a
syllabus or a
curriculum and even as an interactive to
you know it will not get bored with you
asking to explain Concepts again and
again or using different metaphors or
making it applicable making it like this
poetry format putting it in this
narrative context and so even asking how

do I start with this or can you help me understand what I don't know that puts such a centrality and and highlights metacognition in learning and then the project proposals each of them structured according to essentially what a grant officer would be looking for and what a what a project manager would be thinking about in terms of small medium and large projects so this unprecedented High throughput way to fuse even scanty input from a participant in a inclusive open science setting into real projects that if enough people think hey that looks kind of interesting then we're well down the road of actually making it happen using different kinds of of funding which is something Michael lennin mentioned a little bit earlier so it's sort of like we can get really detailed and Technical and Architectural thinking and and and finessing the differences between reinforcement learning deep learning active inference and we saw slides to that like when Carl showed the action perception Loop similarities and differences okay active inference doesn't have a reward function it's not optimizing based upon this trained auxillary reward concept it's just doing something direct with likelihoods like that is critical that is what makes the active inference generative models what they are but then looking at something like this which

wasn't so accessible one to several
years

ago that just feels like opening up and
unleashing what people can do and about
using these as Tools in our

Niche

and I think it's a total open and
unknown what happens when you can get to
level of project and learning plan for
for an arbitrarily complex field
with the patience of an AI That's just
going to accept questions all day long
like that's almost the
bigger applications

question again just zooming out from how
the architectures and the parameters are
similar and different to how are we
going to do what is authentic and right
and best given the tools we have there's
a bigger picture Beyond just holding the
magnifying glass to different
computational

artifacts that's well said Daniel I
really agree it's

transformative if anyone has questions
in the chat or here or any other things
they want
to mention

yeah what's going to happen in the next
two sessions John or what would be your
goals or preferences for for this Arc of
the Symposium for you yeah so I I I
think is as Andrew alluded um this the
session tomorrow is going to be really
on the application of of uh of active
agents uh to a a design of a economic
agent new kind of firm biofirm that's
homeostatic um and the objective of that

is to develop a new kind of incentive structures a point to Collective action um that are based upon mimetic principles that I think we'll see address a lot of the issues that people are having in the whole idea of bi Regional Finance and how to deal with climate change and how to create those incentives so there one of the things I'll kick off with I've been talking to a number of people in the whole fields of of um um regenerative Finance bi where they go bofi um this it ties into the work that was done in Elanor ostrom's work on uh how common pole resource how do you manage a common pole resource how do you incentivize people different scales I really think that and what you identify so the requirements that people are looking for it in order to address these problems the scale and I think what we're be talking about is through the implementation of active inference uh principles into an economic agent that it provides I think a a a reasonable way of of addressing these issues um and that um um and and so that's part of the conversation that we want to have and and and bring together these two these two worlds um then the third session is to really look at this um from a a broader point of view is saying where is this going and this are the people that we have in the panels are very off very much involved with Enterprises um applying it to nonprofits to applying it to S problems looking at what they can

do with llms currently and what the
certain current agentic architectures
are and what happens when when you s
what happens when you go to full
autonomy what does it mean to have
autonomy um in in a in a in a biological
sense something keeping alive and what
does it mean to build systems that have
autonomy this issue of autonomy is sort
of fundamental uh when when when people
think of of autonomous organiz
autonomous cars or decentralized
autonomous organizations or the whole
idea of U the Central Bank the the the
the the principles behind something like
a Bitcoin is is like oh that that that
is autonomous but is it really
autonomous what is this new notion of
autonomy if we base it on biological
principles
um and I I think that's that opens the
door to a very different set of
discussions than than we're currently
having and it also opens the door to new
Notions of governance and how we can
build governance in to maintain the
homeotic states um and Andrew is
alluding to that so I mean that that
that that that's where I become a an
optimist and I'm not an optimist around
the extension of llms and and open Ai
and the and the tech
the tech Giants but I am an optimist in
being able to apply these principles to
to some of the fundamental issues we
having around climate and social equity
and economic
equity awesome okay in the closing

minutes if anyone has lost comments and David's question too if you want to address it Andrew or anyone else any thoughts about how we start to develop a multiscale approach to ethics um I want allow time for other people so I'll briefly say uh so so tomorrow um you know we'll we'll we'll present uh various aspects of the biofirm there will be a little bit of a Redux to some of the things I went through today just to kind of thread The Narrative um and I'll kind of talk about you know homoeconomicus and all these kinds of uh assumptions about human behavior and things like that but uh um I mean multiscale governance I what we're trying to do with our bio firm is to uh you know and clippinger refer to um uh Elanor ostrom's work on the commons and so I I'm going to address that a little bit too but the idea is to be able to have a kind of like adaptable system of multiple agents uh you know and so we have a kind of small Society so to speak and uh you know they're kind of governing the commons around them and by being able to have something like the capacity for one shared goals uh and two adaptive learning such that they can actually adapt what they do and what they believe and actions they carry out have those be localized to the current context which is an essential part of ost's like work is like it's not so simple to just give a singular top down instruction that applies in all places I mean to be able to have that kind of

adaptivity and awareness of the local um
you know while at the same time having
agents who are kind of synced up in
these various ways I think that's a one
way of trying to approach governance and
then from there you might be able to
have
kind of layered you know you could have
individual hubs you know and then you
could have some kind of like
interconnecting system between them that
that extends that to an even broader
scale of multiple hubs and interaction
between them so just some just some
breadcrumbs there um
thanks yeah Ma and then John with the
last word on this
session yeah so I mean the question of
Ethics is a bit it's obviously
complicated but the
the the key part about whether we
coordinate around multiscale goal
sharing Etc share potentiation that's
intrinsic to poweractive inference right
it just is the question that isn't going
to be solved is what is good uh because
you can always say well you know what is
good is good by virtue of being
contextually sort of stable but there
may be other ways to be stable or that
might prioritize other kind of
individuals or entities Etc so I don't
think we're going to solve that question
however what we can solve is the is the
global versus local coherence of certain
kinds of beliefs relative to what might
constitute goodness at given scales and
find where the friction points might

arise because this version of goodness is incompatible with this version of goodness and therefore Uh something's going to give somewhere an error is going to have to propagate so I think that's going to be answered but the the larger broader question you you'd have to look into different schools of thought around what constitutes uh good because we don't necessarily want to go towards um moral realism um because there's a lot of things that are baked into that uh but you might want to go into the uh ethical implications of neomaterialism and how it is about interactions and locality and at that point it becomes about how do certain pockets of ethical framework sort of form and allow to just coexist away from other pockets and what happens when they start interacting uh which again we may be seeing right now in the US yes good point John any last comments on the session oh this is great um and I think U we we now surfacing some of the issues that are really important to to talk about and and um I think Le least well we're going to be presenting is is a is a framework um that at least we can have these discussions in a more rigorous way and in testable way um and uh I think that's part of the way of moving it forward so yeah awesome thank you okay we'll take a one minute break see you soon okay thank you e e